

FIELD PROGRAMME RAHIM – WEEK 5

Time-Series Data, Anomaly Detection & Feature Engineering

By Friday, you should be able to:

- Understand what time-series data is.
- Understand how sensor readings change over time.
- Analyze motor data chronologically.
- Identify trends and abnormal behavior in sensor readings.
- Understand correlations between different sensor measurements.
- Create new features from raw sensor data.
- Understand rolling averages and moving windows.
- Understand the difference between a normal reading and an anomaly.
- Prepare time-based sensor data for Machine Learning.
- Explain what information could be useful for predicting a future motor failure.

Day I – Understanding Time-Series Data

Learn

Study:

- What is time-series data?
- Timestamp
- Sequence of events
- Trends
- Sudden changes
- Peaks and drops
- Noise
- Sensor readings over time

Understand that motor data isn't just a collection of independent rows.

For example:

Time	Temperature	RPM
09:00	60	1480
09:01	61	1479
09:02	62	1478
09:03	64	1475
09:04	68	1460
09:05	74	1420

The important information is not only:

"Temperature = 74"

but also:

"Temperature has been continuously increasing while RPM has been decreasing."

That pattern could potentially be important.

Practical Work

Use the motor dataset from Week 4.

If it does not contain a useful timestamp/time column, create a simulated time column for this exercise.

Program 1 – Load & Sort Sensor Data

Use Pandas to:

- Load the dataset.
- Convert the time column into a datetime format.
- Sort records chronologically.
- Display the earliest and latest readings.

Program 2 – Sensor Trends

Calculate:

- Average temperature per time period.
- Average RPM per time period.
- Average current per time period.

Program 3 – Identify Changes

Find:

- Largest temperature increase.
 - Largest RPM decrease.
 - Largest current increase.
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Questions

Write short answers:

1. Which sensor changes the most over time?
2. Which sensor appears to be the most stable?
3. Are there periods where several sensors change at the same time?
4. Why might the sequence of readings matter when monitoring a motor?

Save as:

time_series_notes.txt

Git Task

Commit and push.

Day 1 - Time Series Analysis

Day 2 – Rolling Averages & Sensor Trends

Today we introduce an important technique used when working with noisy sensor data.

Learn

Study:

- Noise
- Moving average
- Rolling average
- Rolling window
- Smoothing data

Imagine the temperature readings are:

60
61
59
63
60
64
62

The readings jump around.

A rolling average can help us see the **overall trend** rather than reacting to every tiny fluctuation.

Practical Work

Using Pandas:

Create a **5-reading rolling average** for:

- Temperature
- Current
- RPM

For example:

Temperature

↓

Raw readings

↓

Rolling Average

Program I – Temperature Trend

Create a graph containing:

- Original temperature
- Rolling average temperature

The purpose is to visually compare:

Raw sensor data vs smoothed sensor data.

Program 2 – RPM Trend

Do the same for RPM.

Program 3 – Identify Increasing Trends

Try to identify periods where:

- Temperature is increasing.
- RPM is decreasing.
- Current is increasing.

Think About It

Ask:

If the temperature slowly increases for 30 minutes, is that more useful information than one single high temperature reading?

Write your reasoning in:

rolling_average_notes.txt

Git Task

Commit and push.

Day 3 – Correlation Between Sensors

Now we start asking a more interesting question:

Do different sensor measurements appear to be related?

For example:

Temperature ↑

+

Current ↑

↓

Could indicate increased motor load

This doesn't prove that one causes the other, but it may reveal useful patterns.

Learn

Study:

- Correlation
- Positive correlation
- Negative correlation
- No correlation
- Correlation matrix
- Scatter plots

A simple interpretation:

Positive correlation

When A increases, B tends to increase.

Negative correlation

When A increases, B tends to decrease.

No/weak correlation

There isn't an obvious relationship.

Practical Work

Program 1 – Correlation Matrix

Using Pandas, calculate the correlation between:

- Temperature
- Voltage
- Current
- RPM
- Torque
- Vibration

Only use columns that actually exist in his dataset.

Program 2 – Visualize Correlations

Create a correlation heatmap using Matplotlib.

Identify:

- Strong positive relationships.
- Strong negative relationships.
- Weak relationships.

Program 3 – Scatter Plots

Choose two potentially interesting relationships.

For example:

Temperature vs RPM

and

Current vs Temperature

Create scatter plots.

Questions

Answer:

1. Which two sensors have the strongest positive relationship?
 2. Which two have the strongest negative relationship?
 3. Does correlation prove that one sensor causes another to change?
 4. Which relationships might be useful for predicting motor failure?
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Important Concept

Make sure he understands:

Correlation does not mean causation.

This is particularly important because eventually he will be presenting this as an engineering/AI project.

Git Task

Commit and push.

Day 3 - Sensor Correlation Analysis

Day 4 – Anomaly Detection

This is probably the most important day of Week 5.

Learn

Study:

- What is an anomaly?
- Normal behavior
- Abnormal behavior
- Outliers
- Threshold-based anomaly detection
- Statistical anomaly detection

Explain the difference between:

Normal

Temperature:

58

59

60

61

60

59

and:

Potential anomaly

58

59

60

61

92

60

59

The 92°C reading is very different from the surrounding readings.

Practical Work

Program 1 – Simple Threshold Detection

Create rules such as:

Temperature > X → Temperature Anomaly

RPM < Y → RPM Anomaly

Current > Z → Current Anomaly

Use appropriate thresholds for the dataset.

Remember:

These thresholds are for learning purposes unless they come from actual motor specifications.

Program 2 – Statistical Detection

Introduce the idea of:

Mean + Standard Deviation

Identify values that are unusually far away from the normal behavior of the dataset.

You don't need to make him memorize the mathematics.

The important idea is:

"If most readings are around a normal range and one reading is extremely far away, we can flag it as potentially abnormal."

Program 3 – Create an Anomaly Column

Add a new column:

Anomaly

Example:

Temperature	RPM	Current	Anomaly
60	1480	8.2	No
61	1479	8.3	No
82	1310	11.2	Yes
63	1475	8.5	No

Visualization

Create a graph showing:

- Sensor readings
 - Highlighted abnormal readings
-

Questions

Write:

1. What makes a reading an anomaly?
2. Can every anomaly mean that the motor is failing?
3. What other reasons could cause an unusual reading?
4. Why could anomaly detection be useful in predictive maintenance?

Git Task

Commit and push.

Day 5 – Feature Engineering for Motor Failure

Today we introduce an important Machine Learning concept:

Feature Engineering

Explain:

Raw sensor data isn't always the best information to give to a Machine Learning model.

We can create **new features** from the existing data.

For example:

Raw:

Temperature

Could become:

Current Temperature

Average Temperature

Temperature Change

Maximum Temperature

These can potentially give a model more useful information.

Practical Work

Using the motor dataset, create new features.

Feature 1 – Temperature Change

Calculate how much temperature changed from the previous reading.

Example:

Temperature

60

61

63

67

becomes:

Temperature Change

-

+1

+2

+4

Feature 2 – RPM Change

Calculate:

Current RPM - Previous RPM

Feature 3 – Rolling Temperature

Create:

5-reading average temperature

Feature 4 – Rolling Current

Create:

5-reading average current

Feature 5 – Anomaly Flag

Use the anomaly detection logic from Day 4.

Create:

Anomaly = 0

or

Anomaly = 1

Final Mini Project

Motor Behavior Analysis

Create:

`motor_behavior_analysis.py`

or preferably:

`motor_behavior_analysis.ipynb`

Your project should:

1. Load the dataset

`motor_data.csv`

2. Clean the data

3. Sort by time

4. Analyze sensor trends

5. Calculate rolling averages

6. Calculate correlations

7. Detect anomalies

8. Create new features

9. Create graphs

At minimum:

Temperature over time

RPM over time

Current over time

Sensor correlation

Anomalies over time

Final Analysis

At the end, answer:

Question 1

What does normal motor behavior look like?

Question 2

What abnormal behavior did you identify?

Question 3

Which sensors appear to be related?

Question 4

Which sensors appear to be most useful for identifying abnormal behavior?

Question 5

What new features did you create?

Question 6

If you were building a Machine Learning model to predict failure, which features would you use?

Question 7

What additional information would you need to make the prediction more reliable?